Analysis of the erosive effect of different dietary substances and medications.

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Excessive consumption of acidic drinks and foods contributes to tooth erosion. The aims of the present in vitro study were twofold: (1) to assess the erosive potential of different dietary substances and medications; (2) to determine the chemical properties with an impact on the erosive potential. We selected sixty agents: soft drinks, an energy drink, sports drinks, alcoholic drinks, juice, fruit, mineral water, yogurt, tea, coffee, salad dressing and medications. The erosive potential of the tested agents was quantified as the changes in surface hardness (ΔSH) of enamel specimens within the first 2 min ($\Delta SH_{2-0} = SH_{2\,min} - SH_{baseline}$) and the second 2 min exposure ($\Delta SH_{4-2} = SH_{4\,min} - SH_{2\,min}$). To characterise these agents, various chemical properties, e.g. pH, concentrations of Ca, Pi and F, titratable acidity to pH 7·0 and buffering capacity at the original pH value (β), as well as degree of saturation (pK - pI) with respect to hydroxyapatite (HAP) and fluorapatite (FAP), were determined. Erosive challenge caused a statistically significant reduction in SH for all agents except for coffee, some medications and alcoholic drinks, and non-flavoured mineral waters, teas and yogurts ($P < 0{\cdot}01$). By multiple linear regression analysis, 52 % of the variation in ΔSH after 2 min and 61 % after 4 min immersion were explained by pH, β and concentrations of F and Ca ($P < 0{\cdot}05$). pH was the variable with the highest impact in multiple regression and bivariate correlation analyses. Furthermore, a high bivariate correlation was also obtained between $(pK - pI)_{HAP}$, $(pK - pI)_{FAP}$ and ΔSH.